



Wind-powered active water supply enables multilayer solar steam generation for efficient seawater desalination

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HIGHLIGHTS

- Wind-driven mechanical pumping breaks capillary limitations in SSG, ensuring continuous seawater supply.
- Stable seawater fluxes up to 25 L h⁻¹ enables effective multilayer evaporation (~30 cm height).
- Multilayer configuration achieves 1.66 L m⁻² h⁻¹ evaporation, ~1.98× higher than single-layer systems.
- Energy-autonomous and scalable design supports large-area solar-wind desalination in coastal and island settings.

ARTICLE INFO

Keywords:

Solar steam generation
Wind-assisted water pumping
Multilayer evaporation
Active water supply
Evaporation rate

ABSTRACT

Freshwater scarcity remains a critical challenge in coastal and island regions where centralized desalination infrastructure is often limited. Solar steam generation (SSG) has emerged as a promising low-energy approach for seawater desalination; however, most SSG systems rely on passive capillary transport, which restricts water supply and limits the implementation of multilayer evaporation architectures. Here, a hybrid solar-wind-driven multilayer SSG system with active water supply is developed to overcome this limitation. Wind energy directly drives a piston pump that continuously delivers seawater to vertically stacked photothermal membranes composed of carbon spheres embedded in a poly(vinyl alcohol) matrix. The pumping unit operates under typical marine wind speeds (3–7 m s⁻¹), providing a stable water supply of up to 25 kg h⁻¹ and lifting seawater to approximately 30 cm. The active water supply enables uniform wetting of multilayer evaporators and improves solar energy utilization, resulting in an evaporation rate exceeding 1.66 kg m⁻² h⁻¹ under 1 sun illumination, corresponding to a 1.98-fold enhancement compared with a single-layer system. Notably, the solar-to-vapor conversion efficiency reaches ~112% for the multilayer configuration, significantly higher than ~56.7% for the single-layer system, indicating effective utilization of both solar irradiation and additional environmental energy. Stable performance is maintained under oblique irradiation due to interlayer light recycling and improved thermal distribution. This hybrid strategy provides a scalable and energy-autonomous pathway for solar-driven seawater desalination in coastal environments.

1. Introduction

Freshwater scarcity is a growing global challenge, particularly in coastal and island regions where natural freshwater resources are limited while demand continues to increase [1]. Solar steam generation (SSG) has emerged as a promising approach for desalination and water

purification by directly converting solar energy into localized heat at the air-water interface [2,3]. This interfacial heating strategy minimizes thermal losses and enables efficient evaporation under natural sunlight [4], offering clear advantages over conventional desalination technologies in terms of cost, environmental impact, and off-grid applicability [5–7]. Recent advances in photothermal materials have enabled

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<https://doi.org/10.1016/j.desal.2026.120227>

Received 10 March 2026; Received in revised form 20 April 2026; Accepted 25 April 2026

Available online 3 May 2026

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evaporation efficiencies exceeding 90% under one-sun illumination [8–11]. Among the photothermal materials developed for SSG, biomass-derived carbons and conjugated polymers have emerged as particularly attractive candidates owing to their abundance, low cost, and strong broadband solar absorption. Carbonized biomass substrates provide inherently porous, hydrophilic architectures that simultaneously facilitate light harvesting and capillary water transport [12,13]. Conducting polymers such as polyaniline (PANI), polypyrrole (PPy), and polydopamine (PDA) offer tunable optical absorption in the near-infrared region and excellent processability, enabling their integration into diverse evaporator configurations, including foams, membranes, aerogels, hydrogels, and woven fabrics [14–19]. Nevertheless, despite these material advances, a critical but often overlooked limitation lies in the fundamentally passive nature of water supply in conventional SSG systems [17,20,21].

To date, most reported SSG designs rely on passive capillary-driven or direct-contact water transport, which is inherently effective only for planar or single-layer configurations [22–24]. Although significant progress has been achieved through material and structural optimization [7,25,26], these advances do not fundamentally address the limitation of water delivery in vertically scalable architectures. Importantly, this limitation is not a matter of material optimization but a fundamental constraint imposed by passive transport physics. For example, Sajjadi-zadeh et al. [27] reported bio-inspired interfacial solar steam generation systems based on FeNi nanoparticle-coated wood sponge and pine cone substrates. Benefiting from their intrinsic three-dimensional porosity and high hydrophilicity, these substrates enable efficient capillary-driven water transport, while FeNi nanoparticles enhance broadband light absorption and photothermal conversion. As a result, the pine cone-based device achieved a high evaporation flux of $1.73 \text{ kg} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ with an efficiency of 97.59% under one-sun illumination. Similarly, Wang et al. [28] demonstrated a solar-thermal desalination system employing spectrally selective absorbers to maximize solar absorption while suppressing thermal emission losses. By integrating the absorber with wicking materials, the system achieved an overall efficiency of $\sim 73\%$ under one sun and a $\sim 10\%$ enhancement in evaporation rate compared to conventional graphite absorbers.

Despite their distinct material strategies and performance metrics, both representative studies fundamentally depend on a passive capillary-driven water supply. Such mechanisms are intrinsically limited by capillary height and pore connectivity [29,30], rendering them unsuitable for vertically stacked or multilayer SSG architectures. In these configurations, insufficient water delivery may lead to non-uniform wetting, increased thermal losses, and performance degradation. Consequently, the absence of an active and scalable water-supply mechanism, rather than photothermal inefficiency, remains the critical bottleneck preventing the practical realization of multilayer SSG systems.

Coastal and island environments offer abundant renewable energy resources beyond solar irradiation, particularly stable wind energy [31]. In contrast to capillary transport, wind energy enables active, directional, and scalable water delivery, offering a fundamentally different paradigm for SSG system design. Yet, current hybrid solar-wind technologies predominantly focus on electricity generation or thermal enhancement [32], without addressing the core limitation of water transport. As a result, the potential of wind energy to redefine the water supply mechanism in SSG systems has remained largely unexplored.

In this work, we introduce a conceptual shift from passive capillary-driven water supply to wind-driven active water delivery, representing a fundamentally different interpretation of how SSG systems should be designed. Wind energy is directly converted into mechanical motion to pump seawater and continuously feed multiple vertically stacked evaporation layers. This approach eliminates the reliance on capillary forces and enables a controllable, sustained, and scalable water supply, which is unattainable in traditional designs. Carbon sphere-based photothermal membranes embedded in a poly(vinyl alcohol) matrix are

arranged in a multilayer configuration to enhance solar utilization through interlayer light recycling. To the best of our knowledge, this work constitutes a pioneering experimental effort to integrate wind-assisted active water supply into multilayer SSG architectures, enabling scalable and energy-autonomous freshwater harvesting.

2. Experimental section

2.1. Fabrication of the multilayer water harvesting system

The freshwater harvesting system consists of three main components (Fig. 1a):

- (1) A wind-powered mechanical water pumping unit.
- (2) A multilayer photothermal evaporation module.
- (3) A condensation and freshwater collection unit.

2.1.1. Wind-powered mechanical pumping unit

The wind-driven pumping unit comprises a wind rotor, turbine, crank disk, connecting rods, piston, and vertical water pipe (Fig. 1b). The wind rotor has a diameter of 40 cm and consists of 15 blades. The rotor shaft is mechanically coupled to a circular crank disk with a diameter of 20 cm. A connecting rod (20 cm in length) is fixed at the edge of the crank disk to convert rotational motion into linear displacement. During the laboratory experiments, the electric fan was positioned such that the airflow direction was horizontal and aligned with the rotor axis, representing the optimal configuration for maximum

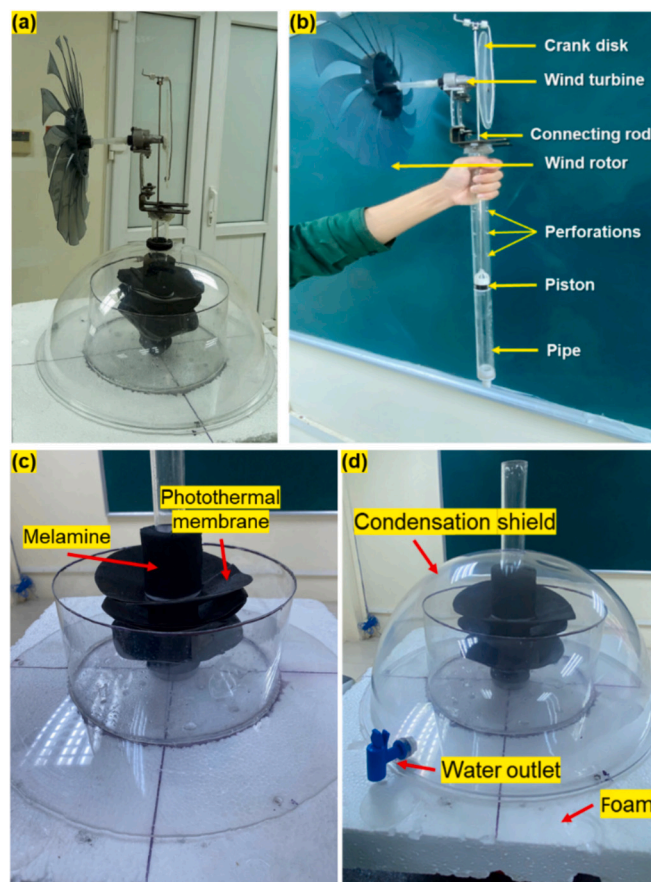


Fig. 1. (a) Photograph of the multilayer water harvesting system. (b) Components of the water pumping system. (c) Photograph of the multilayer photothermal conversion membranes. (d) Photograph of the freshwater collection unit.

torque generation.

The crank disk is connected to a vertical push rod (50 cm in length), which drives a piston housed inside a cylindrical water pipe with an inner diameter of 26 mm and a length of 50 cm. When exposed to wind, the rotor generates rotational motion that is transmitted through the turbine to the crank disk. The resulting reciprocating motion of the piston creates periodic suction, enabling seawater to be actively pumped upward through the pipe. Multiple perforations are introduced along the pipe at predefined positions to distribute water uniformly to different photothermal evaporation layers.

2.1.2. Fabrication of photothermal evaporation membranes

Photothermal membranes were fabricated using carbon spheres (CSs) with an average diameter of $6.64\ \mu\text{m}$, which have been successfully investigated in our previous work [33]. Specifically, 0.1 g of CSs were dispersed into 10 mL of 10 wt% PVA solution, corresponding to approximately 1.0 g of dry PVA. After homogeneous mixing, the suspension was uniformly cast onto a cellulose paper substrate with a diameter of 20 cm and dried at $50\ ^\circ\text{C}$. After drying, the thickness of the obtained PVA@CSs membrane was approximately $50.3\ \mu\text{m}$.

Each membrane was folded into a conical geometry with a base diameter of 15 cm and a height of 5 cm (Fig. 1c). For the multilayer configuration, three identical membranes were vertically stacked with an interlayer spacing of 5 cm (Fig. 3b). Melamine foam spacers were placed between adjacent layers to provide mechanical support and temporary water storage, ensuring a continuous water supply during transient low-wind conditions. A single-layer SSG system consisting of one photothermal membrane and one melamine spacer was fabricated as a reference for performance comparison (Fig. 3c).

2.1.3. Condensation and freshwater collection unit

The condensation unit consists of a transparent plastic dome with a diameter of 40 cm, positioned above the evaporation module (Fig. 1d). The dome is supported by a foam base with dimensions of $5 \times 50 \times 50$ cm. During operation, water vapor generated at the photothermal membranes condenses on the inner surface of the dome and flows downward under gravity. The condensed freshwater is continuously collected through an outlet installed at the bottom of the dome.

2.2. The evaporation process

Fig. 2 illustrates the working principle of the integrated solar-wind steam generation system for water harvesting. Wind energy is first captured by a turbine (Fig. 1b), which converts the kinetic energy of airflow into rotational motion. This mechanical motion drives a piston mechanism that produces periodic reciprocating movement, enabling active pumping of seawater toward the evaporation unit. The pumped water is subsequently delivered to the photothermal layers via

perforations (Fig. 1b), where incident solar irradiation is efficiently absorbed and converted into localized heat. The resulting interfacial heating accelerates water evaporation and generates vapor above the photothermal surface. The produced vapor then rises and condenses on a condensation shield positioned above the evaporation region (Fig. 1d), where the condensed droplets are collected as purified water.

2.3. Characterizations

Solar irradiation was simulated using a xenon lamp system calibrated to a standard intensity of 1 sun ($1\ \text{kW}\cdot\text{m}^{-2}$). During each experiment, the generated water vapor was collected over a fixed time interval, and the mass of the condensed freshwater was measured using an electronic balance with a precision of $\pm 0.0001\ \text{g}$. The evaporation rate was calculated based on the measured mass change and the effective evaporation area. The surface temperature distribution of the photothermal membranes was recorded using an infrared thermal imaging camera (TESTO 870).

Artificial wind was generated using an electric fan (Toshiba F-LSA20 (H)VN, 60 W) with adjustable wind speed settings to simulate realistic marine wind conditions. The wind velocity at the system inlet was monitored and recorded using a thermo-anemometer (EXTECH AN100). All wind speed measurements were performed after stabilization to ensure steady-state airflow during data acquisition.

3. Results and discussion

3.1. Wind-powered water pumping performance

The pumping process under wind-driven operation was visually recorded (Supporting Video S1). The pumping height is a critical parameter for enabling a multilayer SSG architecture, since passive capillary forces are inherently insufficient to transport water to elevated layers, necessitating active water-lifting mechanisms. The wind-driven piston pump successfully lifted water to a height of approximately 30 cm (Fig. 3a), sufficient to supply multiple vertically stacked evaporation layers simultaneously. This result demonstrates that wind-powered mechanical pumping provides a robust and scalable solution to overcome the intrinsic limitations of passive capillary-driven water supply in conventional SSG systems.

The performance of the wind-driven pumping unit was systematically evaluated under wind speeds ranging from 1 to $7\ \text{m}\cdot\text{s}^{-1}$ to assess its feasibility under realistic coastal and island conditions. As shown in Fig. 3b, the pumping system remained inactive at wind speeds below $3\ \text{m}\cdot\text{s}^{-1}$ due to insufficient mechanical torque to overcome the hydrostatic pressure associated with the water column and the relatively large inner diameter of the water pipe (26 mm).

Once the wind speed reached $3\ \text{m}\cdot\text{s}^{-1}$, the system became fully

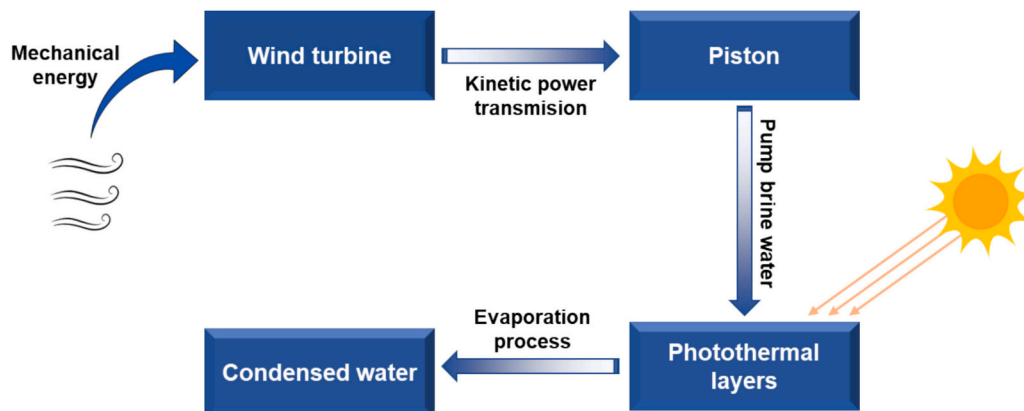


Fig. 2. Schematic illustration of the working principle of the solar-wind steam generation system.

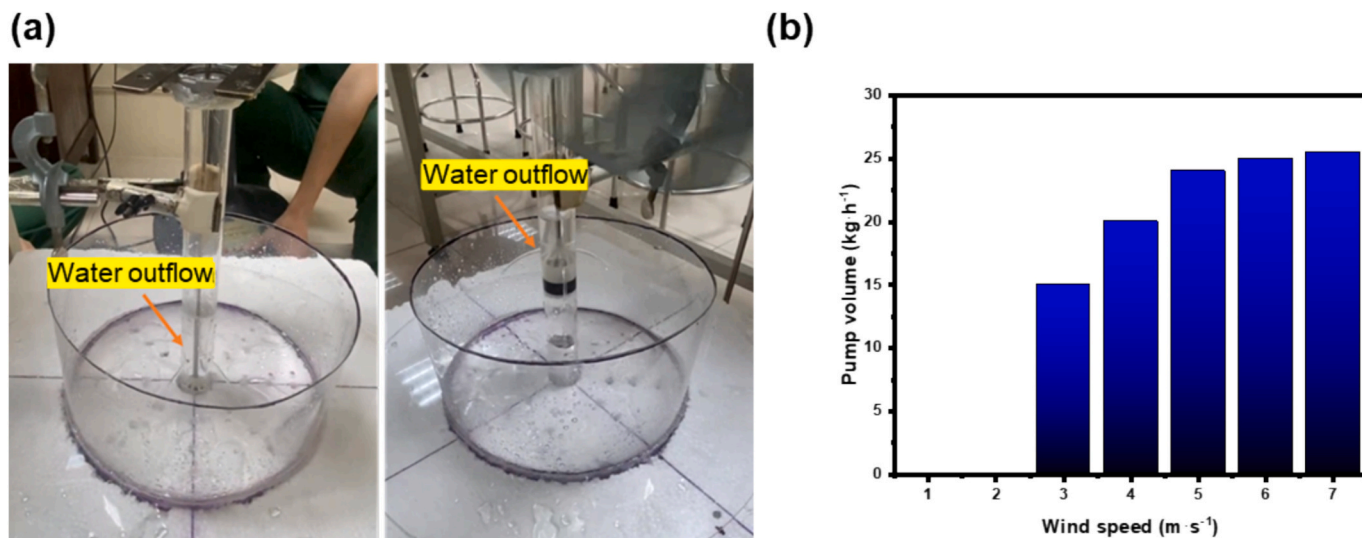


Fig. 3. (a) Water outflow during wind rotor rotation. (b) Pumped water volume as a function of wind speed.

operational, delivering approximately 15 kg·h⁻¹ of seawater. With increasing wind speed, the pumping rate gradually increased and reached a maximum value of approximately 25 kg·h⁻¹ at 7 m·s⁻¹. Notably, wind speeds of 5–6 m·s⁻¹ are commonly observed in coastal and island regions, indicating that the pumping system can operate efficiently under typical field conditions without auxiliary energy input.

A corresponding analysis of the energy conversion efficiency, presented in the Supporting Information, further supports the feasibility of this wind-driven pumping strategy. The calculated results show that the conversion efficiency from wind energy to water supply energy is approximately 0.02%, indicating that the hydraulic energy required for water delivery is several orders of magnitude lower than the theoretical wind power flux intercepted by the rotor. This implies that only a very small fraction of the available wind energy is sufficient to sustain continuous seawater supply to the multilayer evaporator. Consequently, the wind subsystem operates in a low-power regime, where its primary role is not to maximize energy conversion efficiency, but to provide a stable, continuous, and reliable water supply that enables consistent evaporation performance under realistic environmental conditions.

3.2. Water distribution and wetting behavior in multilayer membranes

Uniform water distribution across all evaporation layers is essential for maintaining stable evaporation and preventing localized over-heating. Fig. 4 illustrates the time-dependent wetting process of the multilayer photothermal membranes following activation of the wind-driven pump.

Although the top evaporation layer was positioned approximately 20 cm above the water source, complete wetting was achieved within 300 s. This observation confirms that the active pumping mechanism effectively compensates for gravitational constraints and ensures continuous water delivery even to elevated layers. In contrast, conventional capillary-fed SSG systems typically experience severe delays or incomplete wetting in vertically stacked configurations, resulting in unstable evaporation performance.

The incorporation of melamine foam spacers between layers further enhances system stability by acting as temporary water reservoirs. This design allows short-term buffering of water supply during intermittent wind fluctuations, ensuring continuous evaporation even under transient low-wind conditions. The combined effects of active pumping and intermediate water retention establish a highly reliable water supply network throughout the multilayer structure.

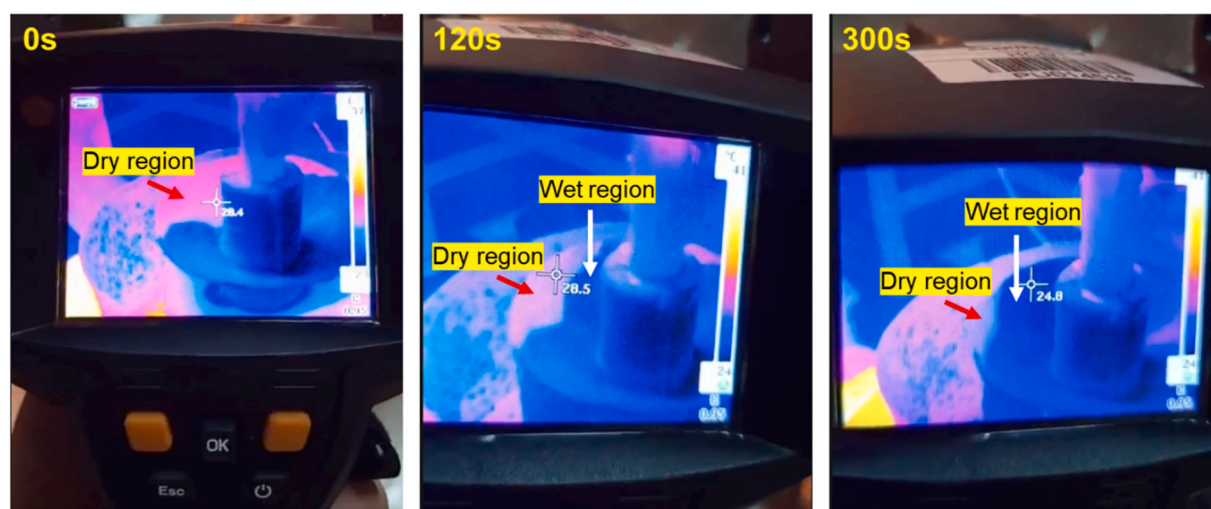


Fig. 4. Time-dependent water infiltration into the evaporation membranes.

3.3. Vapor generation and condensation dynamics

The wettability of the PVA@CSs membrane was evaluated via water contact angle measurements. As illustrated in the time-resolved images (Fig. S1) and supporting Video S2, a water droplet is completely absorbed within 6 s, indicating outstanding hydrophilicity. This rapid water uptake facilitates efficient and continuous water transport to the evaporation interface, which is essential for maintaining high-performance solar steam generation. The evolution of vapor condensation on the transparent dome was monitored to evaluate the dynamic evaporation and condensation processes (Fig. 5). During the initial 5 min of solar irradiation, the condensation surface remained largely transparent, indicating limited vapor accumulation. After approximately 20 min, the inner surface of the dome became visibly fogged, reflecting increased vapor generation from the photothermal membranes.

With prolonged irradiation (60–120 min), discrete water droplets coalesced into larger droplets and flowed downward under gravity, forming continuous water streams toward the collection outlet. This observation confirms that the evaporation rate exceeds the condensation threshold, enabling continuous freshwater harvesting rather than intermittent droplet formation. Importantly, no salt crystallization was observed on the evaporation membranes during operation, suggesting that active water replenishment effectively prevents salt accumulation.

To further validate the practical applicability of the proposed system, an outdoor experiment was conducted under real environmental conditions. As shown in Fig. S3, the integrated floating device maintained stable buoyancy throughout the experiment. Despite suboptimal weather conditions characterized by high ambient relative humidity ($\sim 80\%$) and limited solar irradiance ($\sim 0.4 \text{ kW}\cdot\text{m}^{-2}$), condensed water droplets were clearly visible across the entire inner surface of the dome-shaped cover after only 5 min of operation, providing direct visual evidence of active vapor generation and condensation. These outdoor observations are consistent with the laboratory-scale condensation dynamics described above, and collectively confirm that the system is capable of sustained freshwater harvesting under realistic operating conditions.

3.4. Thermal distribution and heat localization mechanisms

The UV–Vis absorption spectrum (200–850 nm) of the PVA@CSs membrane, presented in Fig. S2, demonstrates consistently high

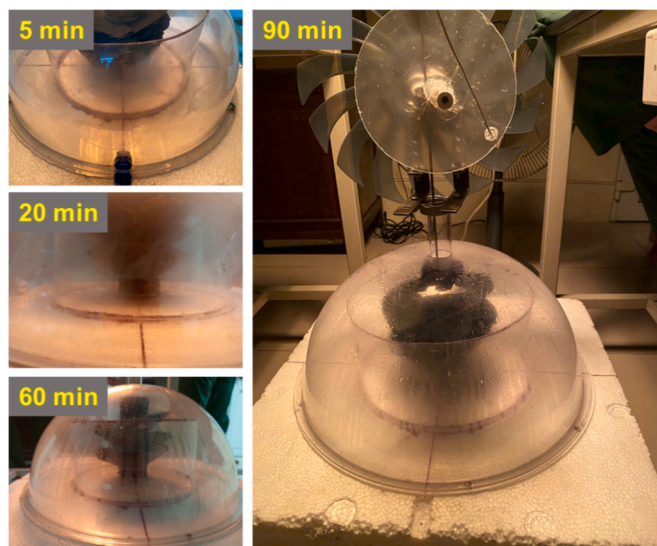


Fig. 5. Time-resolved photographs of water vapor condensation at different time intervals.

absorbance exceeding 92% across the entire measured range, confirming its excellent broad-spectrum light-harvesting capability. Thermal imaging analysis provides further insight into the performance enhancement mechanisms (Fig. 6a). At normal incidence (90°), the top evaporation layer exhibited the highest surface temperature, reaching up to 51.7°C , while lower layers received limited irradiation. As the incident angle decreased to 60° and 30° , the thermal distribution across lower layers became increasingly uniform, with surface temperatures ranging from 39 to 45°C . Notably, at oblique angles, the lower layers exhibited higher and more uniform temperatures than the top layer at normal incidence ($\sim 36^\circ\text{C}$). This observation confirms that the multi-layer configuration effectively redistributes solar energy under non-ideal illumination conditions, enabling sustained evaporation across multiple layers.

To further elucidate the variation in effective illuminated area under different solar incident angles, optical simulations were performed using Blender software (Fig. 6b). The simulated light coverage reveals a strong angle-dependent irradiation behavior of the multilayer architecture. At normal incidence (90°), solar irradiation is largely confined to the top evaporation layer, while the lower layers receive minimal direct illumination, consistent with the thermal distribution observed in Fig. 6a. As the incident angle decreases to 60° , solar radiation becomes more uniformly distributed across all layers, accompanied by a substantial increase in the total illuminated area. This expanded light coverage enables lower layers to actively participate in solar absorption through interlayer reflection and multiple scattering. At a further reduced angle of 30° , the overall light exposure remains significantly higher than that at normal incidence.

Building upon this angle-dependent light distribution (Fig. 6b), the interlayer optical recycling mechanism illustrated in Fig. 6c further enhances solar utilization. Light reflected from the lower evaporation layers can be reabsorbed by the upper layers, leading to multiple reflections within the multilayer structure. This interlayer optical trapping effect effectively extends the optical path length, allowing photons to be repeatedly recycled until absorbed by the photothermal membranes. As

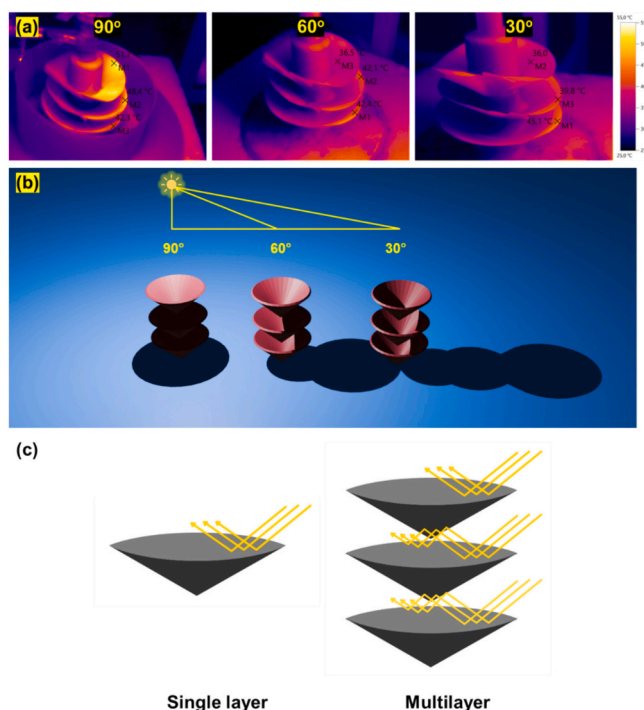


Fig. 6. (a) Temperature distribution of photothermal membranes under different solar incident angles. (b) 3D simulation of light irradiation on the system at different incident angles. (c) Schematic illustration of light reflection in single-layer and multilayer systems at a solar incident angle of 60° .

as a result, reflection losses, commonly observed in single-layer SSG systems, are substantially suppressed, enabling near-complete utilization of incident solar energy. This mechanism is further supported by the quantitative analysis summarized in Table S1. For a vertical pitch of 5 cm between adjacent layers, the total illuminated area increases from 176.7 cm² at 90° to 262.8 cm² at 60°, and further to 421.3 cm² at 30°, corresponding to an approximately 2.4-fold enhancement compared to normal incidence. This result indicates that under oblique irradiation, additional layers become optically accessible and actively participate in light absorption, thereby contributing to the enhanced evaporation rate. In addition, the total evaporation surface area increases proportionally with the number of stacked layers, from 212.4 cm² for a single layer to 637.2 cm² for three layers. This nearly threefold increase in active surface area further amplifies the vapor generation capacity. Overall, the synergy between interlayer optical recycling, enhanced light interception, and enlarged evaporation interface is the key factor responsible for the significantly improved performance of the multilayer system.

3.5. Evaporation rate performance: Multilayer vs. single-layer systems

The evaporation rates of the multilayer and single-layer SSG systems were quantitatively compared under identical environmental conditions (Fig. 7). At a wind speed of 3 m·s⁻¹, the multilayer system achieved a freshwater yield of approximately 1.62 kg·m⁻²·h⁻¹, significantly higher than 0.84 kg·m⁻²·h⁻¹ obtained from the single-layer system. With increasing wind speed, the freshwater yield of the multilayer system increased only slightly, reaching a maximum of 1.66 kg·m⁻²·h⁻¹ at 4 m·s⁻¹, corresponding to a 1.98-fold enhancement relative to the single-layer configuration. Based on these evaporation rates, the corresponding solar-to-vapor conversion efficiencies were estimated to be ~56.7% for the single-layer system and ~112% for the three-layer configuration. The apparent efficiency exceeding 100% indicates that, in addition to direct solar input, the multilayer system effectively utilizes supplementary energy from the environment, such as interlayer light recycling and ambient heat harvesting, thereby demonstrating its superior energy utilization compared to the single-layer design.

The limited sensitivity to further wind speed increase can be attributed to the rapid and complete wetting of all evaporation layers at moderate wind velocities, after which the evaporation process becomes predominantly governed by solar irradiation rather than water supply. These results indicate that the primary role of wind energy in the proposed system is to enable stable multilayer operation, rather than to continuously enhance evaporation rates.

Despite the enhanced evaporation performance, prolonged operation inevitably leads to salt accumulation on the photothermal membrane surface. As shown in Fig. S4, visible salt deposits began to appear at the

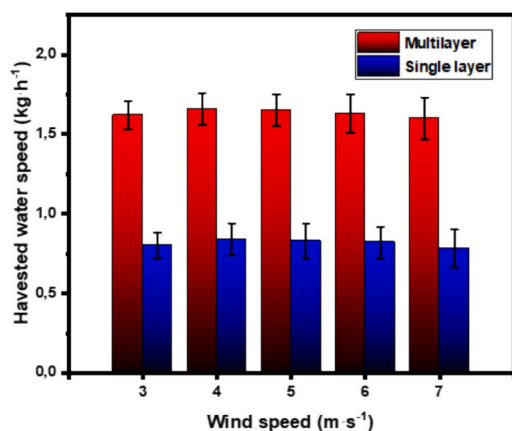


Fig. 7. Comparison of evaporation rate between the multilayer and single-layer SSG systems as a function of wind speed under 1 sun illumination.

membrane surface after more than 24 h of continuous illumination under 3.5 wt% NaCl solution. The continuous wind-driven water supply provides a persistent flushing effect that delays salt crystallization, and the conical structural design facilitates gravitational drainage of concentrated brine away from the active evaporation zone. These combined effects significantly suppress salt buildup during normal operation periods, confirming the practical viability of the proposed system for extended use.

3.6. Effect of solar incident angle on evaporation performance

In real-world applications, solar irradiation rarely occurs at normal incidence throughout the day. Therefore, the adaptability of SSG systems to varying solar angles is a critical performance metric. Fig. 8 compares the evaporation rate at solar incident angles of 90°, 60°, and 30°. For the single-layer system, freshwater yield decreased sharply as the incident angle decreased due to the reduction in effective projected area and direct solar absorption. In contrast, the multilayer system exhibited a non-monotonic trend: evaporation rate increased as the incident angle decreased from 90° to 60°, followed by a moderate decline at 30°. This counterintuitive enhancement at 60° arises from interlayer light reflection and multiple scattering effects inherent to the multilayer architecture, as explained in Fig. 8. Oblique sunlight is reflected between adjacent layers, increasing the effective optical path length and promoting additional absorption that is absent in single-layer systems.

Table 1 compares the evaporation performance of representative photothermal materials and systems reported in previous studies with the solar-wind hybrid system developed in this work. Reported evaporation rates generally fall within the range of 1.42–2.26 kg·m⁻²·h⁻¹, depending on the material composition and structural configuration of the evaporator. For example, a sodium alginate-tannic acid hemispheric evaporator exhibits an evaporation rate of 2.26 kg·m⁻²·h⁻¹, while systems based on pine cone/FeNi, Ti₃C₂T_x MXene, and PGO-Ag//GO achieve evaporation rates of 1.73, 1.60, and 1.59 kg·m⁻²·h⁻¹, respectively. A structure composed of graphene nanoplatelets with chitosan-coated melamine foam shows a slightly lower evaporation rate of 1.42 kg·m⁻²·h⁻¹. The solar-wind hybrid system presented here achieves an evaporation rate of 1.66 kg·m⁻²·h⁻¹, which is comparable to or higher than several previously reported photothermal systems. Notably, this performance is achieved using a relatively simple architecture consisting of only three photothermal layers. The layered design provides a modular platform that can be readily expanded. Increasing the number of photothermal layers is expected to enhance solar absorption and heat localization, thereby further improving the evaporation rate. This

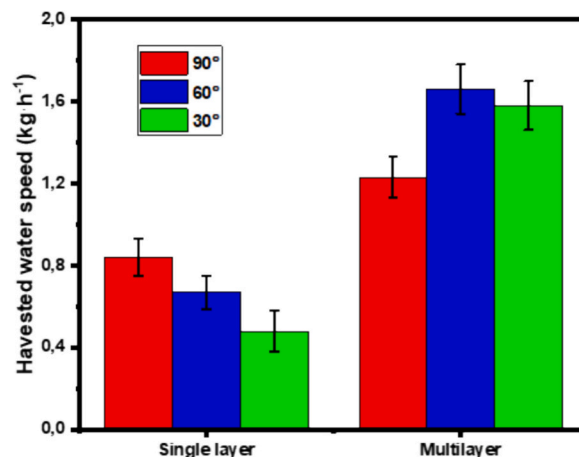


Fig. 8. Comparison of evaporation rate between multilayer and single-layer systems at different wind speeds.

Table 1
Comparison of evaporation rates of representative photothermal materials and systems.

Photothermal materials/ systems	Evaporation rate (kg·m ⁻² ·h ⁻¹)	References
Sodium Alginate-Tannic Acid Hemispheric Evaporator	2.26	[34]
Pinecone/FeNi	1.73	[27]
Ti ₃ C ₂ T _x MXene	1.60	[35]
PGO-Ag//GO	1.59	[36]
Graphene nanoplatelets/chitosan-coated melamine foam	1.42	[37]
Solar-wind hybrid system	1.66	This work

structural flexibility highlights the potential of the proposed architecture as a scalable platform for high-efficiency solar steam generation.

4. Conclusions

A novel hybrid solar-wind-driven multilayer water harvesting system has been successfully developed to address the fundamental water supply limitation of conventional SSG. By directly converting wind energy into mechanical pumping, seawater is actively delivered to vertically stacked photothermal evaporation layers, enabling scalable multilayer operation without electricity or external power sources. The wind-powered pump operates reliably under realistic marine wind conditions (3–7 m·s⁻¹), supplying up to 25 kg·h⁻¹ of water and lifting seawater to heights of ~30 cm. As a result, the multilayer SSG system achieves an evaporation rate exceeding 1.66 kg·m⁻²·h⁻¹, representing a 1.98-fold enhancement compared with a single-layer configuration. Moreover, the multilayer architecture maintains superior performance under oblique solar irradiation due to interlayer light recycling and improved thermal distribution. This study demonstrates a scalable design concept that can be extended to large-scale deployment, enabling the construction of solar-wind-powered water harvesting fields for evaporation rate.

CRedit authorship contribution statement

Duong Duy Son: Conceptualization, Methodology, Software, Writing- Original draft preparation. **Nguyen Thi Bích Ngọc:** Data curation, Writing- Original draft preparation. **Vo Quoc Thanh:** Visualization, Investigation, Data curation. **Le Thanh Tung:** Visualization, Investigation, Data curation. **Bui Hoang Gia Bao:** Software, Validation. **Do Ta Dinh Vinh:** Resources, Data curation. **Pham Nguyen Hoang Tan:** Formal analysis, Investigation. **Nguyen Ba Hung:** Formal analysis, Investigation, Software. **Nguyen Minh Hoang:** Writing- Reviewing and Editing, Supervision. **Phuong Anh Lam:** Methodology, Formal analysis. **Van-Duong Dao:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was financially supported by the Vietnam National Foundation for Science and Technology Development (NAFOSTED) (grant number 103.02-2021.106). The authors also thank the Vietnam Military Medical University for institutional support of this research.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.desal.2026.120227>.

Data availability

Data will be made available on request.

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